Optimal Carbon Tax with Dynamic Skill Heterogeneity*

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Abstract

How should a government design a carbon tax when it is not only concerned by mitigating carbon emissions, but also income redistribution? This article extends a dynamic Mirrlees taxation model, typically used to characterize optimal labor wedges and savings wedges, to study the optimal commodity wedge between two goods, one of which is polluting. First, starting with the basic case where there are two non-polluting goods, I derive the optimal commodity wedge with non-separable preferences in a dynamic environment. Second, in the case where one of the goods is polluting and pollution damages both output and utility, I derive a modified optimal carbon tax whose output damages component remains Pigouvian and whose utility damages component is distorted by the labor wedge as well as the gap in complementarity in utility between each good and labor supply. Third, I derive a third-best commodity wedge when one of the other wedges (labor or savings) cannot be set optimally, mirroring the case when commodity wedges compensate for a suboptimal tax system, as well as a third best labor and savings wedge when the commodity wedge cannot be set optimally. In a third-best setting, the "free" wedges have to compensate for the suboptimal wedge. However, the direction of the adjustment of the "free" wedges depends on the structure of preferences, more specifically the size and sign of cross-derivatives.

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1 Introduction

While there is a growing consensus on the existence of manmade climate change, there is still much disagreement on the appropriate policies to address it, in part due to equity concerns. Although carbon pricing is often lauded by experts as one of the most efficient tools to address climate change, claims that it may unfairly burden lower income households have stirred much public debate on carbon taxation in recent years in countries like France (*gilets jaunes* protests) and Canada (end of consumer carbon tax in early 2025).

One possible reason for this disconnect might be that most integrated assessment models, which are macroeconomic growth models with climate modules used to estimate the optimal level of emission reductions, often view each country or region as a single unit and therefore abstract from between-household inequalities. Hence, governments are guided by analyses underpinned by representative agent integrated assessment models such as DICE, FUND, and PAGE (Environment and Climate Change Canada (2023); Interagency Working Group on Social Cost of Greenhouse Gases (2021)) instead of their heterogeneous agent counterparts, potentially leading to climate policies that do not align with their redistributive goals. Although policy documents such as the sixth IPCC assessment report stress the importance of taking into account within-country inequalities when estimating the optimal price of carbon, they fail to provide precise policy guidance (IPCC 2022). How then should a carbon tax be designed to align with the government's income redistribution goals?

Most research on the distributional aspects of carbon taxation takes an *ex post* perspective, asking whether existing carbon taxes are regressive and comparing the progressivity of different revenue recycling schemes. However, there is as of yet scarce knowledge on if and how income inequalities should affect the optimal design of the carbon tax itself. To this end, I extend a dynamic Mirrlees income taxation model (Golosov, Kocherlakota, et al. 2003, Golosov, Troshkin, et al. 2016, etc.) with two goods, one of which is polluting, to study the optimal joint design of labor income taxes, capital income taxes, and carbon taxes. To my knowledge, this is the first derivation of the optimal carbon tax in a simple dynamic heterogeneous agent integrated assessment model, excluding analyses with multiple regions or multiple generations.

While some theoretical results have been derived in static models with income inequality (Cremer et al. (1998); Jacobs and de Mooij (2015)), they cannot be directly applied to integrated assessment models which are dynamic in nature. Since such models are used by policymakers to quantify the appropriate level of a carbon tax, this is an important first step towards bridging theory and practice.

This research is part of the literature on second-best carbon taxation. A number of studies have shown that introducing an exogenous revenue requirement financed with distortionary income taxation to a representative agent model with a climate externality lowers the optimal level of a carbon tax as environmental quality competes with raising revenue (Goulder (1995); Bovenberg and de Mooij (1994); Barrage (2019)). Such a result may partially explain why the price of carbon in most countries falls short of the value recommended by standard climate-economy models. That is, countries may be reluctant to implement bolder climate policies because they worry about their ability to raise revenue in addition to the usual political-economy reasons.

Douenne et al. (2024) extend the representative-agent Ramsey setting in Barrage 2019 to a heterogeneous-agent Ramsey setting using a methodology developed in Werning (2007) to explore how inequality affects the structure of the optimal carbon tax. Similarly to Barrage, they find that the they find that the optimal carbon tax should be slightly lower than in representative-agent models.

However, all these approaches assume that markets are complete such that agents can perfectly insure themselves against idiosyncratic income shocks, that the planner only has access to linear tax instruments, and that there is no economic rationale for positive capital taxation. In a static environment, Jacobs and de Mooij (2015) shows that a tax on polluting goods needs not be adjusted for the presence of income taxation if utility is weakly separable and the government is free to levy a non-linear labor income tax. Otherwise, if the polluting good is relatively more complementary with leisure, if the polluting good is relatively more demanded by high-ability households, or if the government cannot redistribute as much as desired, the optimal tax on polluting goods may be lower or higher than in a first-best environment. Hence, higher carbon taxes do not necessarily contradict the government's fiscal goals as implied by a representative agent model or even a heterogeneous agent model with linear tax instruments. Whether the results in Jacobs and de Mooij (2015) apply in a dynamic environnement is an open question that will be answered in this paper.

Using a dynamic Mirrlees model of income taxation, I extend Jacobs and de Mooij 2015 to a dynamic environment where agents face uninsurable labor productivity shocks and their consumption generates an externality. In order to separate the role of commodity taxation and of externality correction, I first derive a formula for the optimal commodity tax in section 3. Then, I derive the main equations of a dynamic Mirrlees model with externalities in section 4. Afterwards, I consider in section 5 the case where a wedge is constrained suboptimal and see the implications of this third-best tax system. Finally, I perform simulations to compare the standard representative agent carbon tax to the optimized heterogeneous agent carbon tax in section 6 and I conclude in section 7.

2 Environment

A natural deviation from the standard dynamic Ramsey representative agent framework which explicitly models income heterogeneity and where taxation provides an economic benefit to agents is the dynamic Mirrlees heterogeneous agent framework developed in papers such as Golosov, Kocherlakota, et al. (2003) and Golosov, Troshkin, et al. (2016). I adapt this setup to an environmental context. In order to maintain focus on the effect of interhousehold inequality on the optimal design of the carbon tax, I abstract from other concerns such as international equity or intergenerational equity. As such, I consider a single generation and a single country.

The economy lasts T periods, with T being potentially infinite, and each agent has the following preferences over two goods c and q, labor supply l, and a pollutant stock S.

$$\mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U(c_t, q_t, l_t, S_t) \right]$$

$\beta \in (0, 1)$ is the discount factor. The utility function U is time-separable, but otherwise allows non-separability between its arguments. The good c is a "clean" good while the good q is a "dirty" good generating pollution. Emissions from consuming the dirty good contribute to a persistent pollutant stock S which evolves according to a function of the current aggregate consumption of the polluting good Q and past aggregate consumption of the polluting good.

$$S_t = S\left(Q_0, Q_1, ..., Q_{t-1}, Q_t\right)$$

This characterization of the evolution of the pollutant stock allows for current emissions to affect future pollutant accumulation, for example by having last period's emissions remain in the atmosphere after linear depreciation α.

$$S_t = Q_t + \alpha Q_{t-1}$$

For generality, I do not restrict the form of the pollutant accumulation function. However, in order to be able to more easily rewrite the problem in a recursive form, I assume that emissions have no effect after s periods.

$$S_t = S\left(Q_{t-s}, ..., Q_{t-1}, Q_t\right)$$

Aside from the presence of two goods and for the accumulation of pollutants harming utility, the model follows the basic setup of the dynamic Mirrlees literature. At $t = 0$, agents draw their initial skill (type) θ_0 from a distribution $F_0(\theta)$. Subsequently, future skills are drawn from a Markov process $F_t(\theta|\theta_{t-1})$, meaning the likelihood of drawing a particular skill at time t depends on the skill drawn at time $t-1$ but not on skills drawn previously. At period t, each agent knows their skill draws up to period t, which I denote by a skill history vector $\theta^t = (\theta_0, ..., \theta_t)$. Let Θ^t be the set of all possible skill histories.

An agent of type θ_t who supplies l_t units of labor produces $y_t = D(S_t)\theta_t l_t$ units of output, where $D(S_t)$ is the fraction of output reduced by pollution damages. Whereas total output is observed by everyone, both skill θ_t and labor supply l_t are private information. Moreover, consumption of both goods is also publicly observable[1]. Let $c_t(\theta^t) : \Theta^t \to \mathbb{R}_+$, $q_t(\theta^t) : \Theta^t \to \mathbb{R}_+$, and $y_t(\theta^t) : \Theta^t \to \mathbb{R}_+$ be the agents allocation of the first consumption good, the second consumption good, and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \to \Theta^t$ be a function describing how an agent of type θ_t chooses to report their type and let Σ^t be all possible reporting strategies. Agents can use capital to transfer resources between periods, earning interest $R > 0$. The social welfare function is as follows.

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U(c_t, q_t, l_t, S_t) \right] dF_0(\theta)$$

To simplify exposition, I am supposing an utilitarian planner above. Let U_c, U_q, and U_l denote partial derivatives with respect to c, q, and l. Since $y = D(S)\theta l$, I can write $U(c, q, l)$ as $U\left(c, q, \frac{y}{D(S)\theta}\right)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ.

Any admissible allocation must be incentive-compatible, meaning the agent must weakly prefer the truth-telling strategy over any other possible strategy. For all strategies and skill histories,

$$\mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{D(S_t)\theta_t}\right) \middle| \theta \right] \geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U\left(c_t(\sigma^t(\theta^t)), q_t(\theta^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{D(S_t)\theta_t}\right) \middle| \theta \right]$$

[1] This is a plausible assumption for some industries such as electricity and air travel. For a wider set of goods whose consumption cannot be as easily observed, the planner's optimum will be further constrained and the results in this paper will no longer hold.

Moreover, for all periods, the resource constraints must hold.

$$\mathbb{E}_0[c_t(\theta^t) + q_t(\theta^t)|\theta] + k_{t+1} \leq \mathbb{E}_0[D(S_t)y_t(\theta^t)|\theta] + Rk_t$$

This set of constraints can be consolidated in a single intertemporal constraint.

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^{T} \frac{c_t(\theta^t) + q_t(\theta^t)}{R} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^{T} \frac{D(S_t)y_t(\theta^t)}{R} \middle| \theta \right] dF_0(\theta)$$

Combining the planner's objective as well as the incentive-compatibility and resource-feasibility constraints yields the following planner's problem. For all strategies and skill histories,

$$\max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t,\ t=0,\ldots,T}} \int_0^\infty \left(\mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{D(S_t)\theta_t} \right) \middle| \theta \right] \right) dF_0(\theta)$$

s.t.

$$\mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{D(S_t)\theta_t} \right) \middle| \theta \right] \geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\theta^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{D(S_t)\theta_t} \right) \middle| \theta \right]$$

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^{T} \frac{c_t(\theta^t) + q_t(\theta^t)}{R} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^{T} \frac{D(S_t)y_t(\theta^t)}{R} \middle| \theta \right] dF_0(\theta)$$

$$S_t = S(Q_{t-s}, ..., Q_{t-1}, Q_t)$$

Now that we have setup the problem, we are ready to characterize the optimal joint tax system, which now includes a corrective wedge.

3 Optimal dynamic non-linear commodity wedge

Since a general formula for the optimal commodity wedge in a dynamic Mirrlees environment with non-separable preferences has not yet been derived in the literature, I start by deriving it in an economy with two goods but no pollution. For an agent with history θ^t, I define labor distortions, savings distortions, and commodity distortions as follows.

$$
\begin{aligned}
1-\tau_t^y(\theta^t) &\equiv -\frac{U_l\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right)}{\theta_t U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right)} \\
1-\tau_t^s(\theta^t) &\equiv \frac{1}{\beta R}\frac{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right)}{\mathbb{E}_t\left[U_c\left(c_{t+1}(\theta^{t+1}), q_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}\right)\right]} \\
1+\tau_t^g(\theta^t) &\equiv \frac{U_q\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta^t}\right)}{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta^t}\right)}
\end{aligned}
$$

Moreover, I define the following elasticities

- $\epsilon_t(\theta) \equiv \frac{U_{ll,t}(\theta)l_t(\theta)}{U_{l,t}(\theta)}$: the inverse Frisch elasticity of labor supply
- $\sigma_t(\theta) \equiv -\frac{U_{cc,t}(\theta)c_t(\theta)}{U_{c,t}(\theta)}$: the elasticity of intertemporal substitution
- $\gamma_t(\theta) \equiv \frac{U_{cl,t}(\theta)l_t(\theta)}{U_{c,t}(\theta)}$: the level of complementarity between good c and labor supply
- $\kappa_t(\theta) \equiv \frac{U_{ql,t}(\theta)l_t(\theta)}{U_{q,t}(\theta)}$: the level of complementarity between good q and labor supply

The problem can be written recursively, using the results in Kapička (2013)[2].

$$V_t(\hat{w}, \hat{w}_2, \theta_-) = \min_{c,q,y,u,w,w_2} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta) \right] f_t(\theta|\theta_-) d\theta$$

$$\text{s.t.}$$

$$\dot{u}(\theta) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta} \right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta) f_t(\theta|\theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta$$

$$u(\theta) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta} \right) + \beta w(\theta)$$

Solving the above problem, I find that the structure of the optimal labor income tax and of the optimal capital income tax is unchanged from Golosov, Troshkin, et al. 2016. However, due to the presence of two goods, there is an additional commodity wedge which might be non-zero if utility is non-separable between labor and consumption.

Result 3.0.1 (Optimal commodity wedge)**.**
The optimal commodity wedge with two goods is

$$\frac{\tau_t^g(\theta)}{1 + \tau_t^g(\theta)} = \left(\frac{\tau_t^y(\theta)}{1 - \tau_t^y(\theta)} \right) \frac{\gamma_t(\theta) - \kappa_t(\theta)}{1 + \epsilon_t(\theta) - \gamma_t(\theta)}$$

Proof. See appendix A.1. □

This equation links the optimal commodity wedge to the the labor wedge and the relative complementarity of labor supply between the two goods. If utility is separable between labor supply and both goods, $\gamma_t(\theta) = 0$ and $\kappa_t(\theta) = 0$ such that this wedge is zero.

$$\frac{\tau_t^g(\theta)}{1 + \tau_t^g(\theta)} = 0$$

[2]Kapicka considers that consumption comes from an arbitrary convex and compact set, which means it is not restricted to a scalar and can very well be a vector of multiple consumption goods.

Likewise, the optimal commodity wedge is zero when the labor wedge is zero, that is when the planner has no need to distort labor supply. Overall, the main mechanism behind the optimal commodity wedge present in static frameworks such as Mirrlees (1971) and Christiansen (1984) still applies in a dynamic framework. That is, whether or not there should be a commodity wedge depends on whether either consumption good is more or less complementary with labor supply (i.e. $\gamma \neq \kappa$). For example, the clean good being relatively more complementary with labor supply ($\gamma > \kappa$) motivates a positive tax on the dirty good.

However, unlike a commodity wedge derived in a static environment, this commodity wedge is time-varying and depends not only on an individual's current type θ_t but on their full history of shocks up to θ_t. If we suppose preferences to be such that the elasticities are constant, then it becomes clear that the time path of the optimal commodity wedge is proportional to the time path of the optimal labor wedge. If elasticities are variable, then the time path of the optimal commodity wedge is also modulated by how type affects the relative complementarity of labor supply between the two goods. Although the time path of the optimal commodity wedge is less trivial in this latter case, it remains that the optimal commodity wedge is a skill-dependent and time-variying object in a dynamic model.

4 Optimal dynamic non-linear wedges with externalities

Now let us go back to the case where good c is "clean" and good q is "polluting". The global incentive constraints in the planner's problem can be substituted using the results in Kapička (2013) to yield the following recursive problem.

$$V_t(\hat{w}, \hat{w}_2, \theta_-, Q_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - D(S)y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, Q) \right] f_t(\theta|\theta_-) d\theta$$

$$\text{s.t.}$$

$$\dot{u}(\theta, S) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta) f_t(\theta|\theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta$$

$$u(\theta, S) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right) + \beta w(\theta)$$

$$S = S \left(\cdots, \int q(\theta) f(\theta) d\theta \right)$$

Solving the above problem, the structure of income taxes is once again unchanged. However, the commodity wedge now features what amounts to a modified Pigouvian term where marginal damages of pollution to utility are adjusted for the planner's fiscal concerns. However, marginal damages to production are simply aggregated across types.

Result 4.0.1 (Optimal corrective wedge)**.**
The optimal corrective wedge between the "clean" good and the "polluting" good is

$$\frac{\tau_t^g(\theta) - \Delta_t}{1 + \tau_t^g(\theta)} = \left(\frac{\tau_t^y(\theta)}{1 - \tau_t^y(\theta)} \right) \frac{\gamma_t(\theta) - \kappa_t(\theta)}{1 + \epsilon_t(\theta) - \gamma_t(\theta)}$$

$$\Delta_t \equiv \underbrace{\sum_{s=t}^{T} \int_{\Theta^s} \frac{1}{R^s} \left[D'(S)\theta l \right] S_{T+1-s,s} f_s}_{\textit{Output damages}} + \underbrace{\sum_{s=t}^{T} \int_{\Theta^s} \frac{1}{R^s} \left[1 - \left(\frac{\tau_s^y(\theta)}{1 - \tau_s^y(\theta)} \right) \frac{\gamma_s(\theta) - \xi_s(\theta)}{1 + \epsilon_s(\theta) - \gamma_s(\theta)} \right] \frac{U_{S,s}}{U_{c,s}} S_{T+1-s,s} \; f_s}_{\textit{Utility damages}}$$

Proof. See appendix B.1. □

Although this paper uses a different simpler characterization of climate dynamics, posing a direct link between aggregate dirty good consumption and output and utility, we may still draw a parallel with Barrage 2019 which derives optimal carbon taxes in a dynamic Ramsey model, which features a representative agent and therefore no redistributive motive for taxation. Like equation 3.1 of Barrage (2019), the optimal carbon tax is a sum of a component correcting for output damages and one correcting for utility damages. However, the presence of non-zero labor income taxes and non-zero capital income taxes do not necessarily cause a distortion in the carbon tax as in the dynamic Ramsey framework. While the component correcting for output damages is always Pigouvian in our set up, the component correcting for utility damages is only non-Pigouvian when utility is non-separable and one of the goods is relatively more complementary with labor, incentivizing labor supply. This extends the results in Jacobs and de Mooij 2015 to a dynamic environment. When there are no labor distortions ($\tau^y = 0$) or when neither consumption nor the environment are more complementary with labor supply ($\gamma_s = \xi_s$), the optimal pollution tax is Pigouvian and there is no commodity tax component.

$$\tau_t^g(\theta) = \sum_{s=t}^{T} \int_{\Theta^s} \frac{1}{R^s} \left[D'(S)\theta l\right] S_{T+1-s,s} f_s \; + \; \sum_{s=t}^{T} \int_{\Theta^s} \left(\frac{1}{R^s} \frac{U_{E,s}}{U_{c,s}} S_{T+1-s,s} \right) f_s$$

If $\tau^y \neq 0$ and $\gamma_s \neq \xi_s$ but $\gamma_s = \kappa$,

$$\tau_t^g(\theta) = \sum_{s=t}^{T} \int_{\Theta^s} \frac{1}{R^s} \left[D'(S)\theta l\right] S_{T+1-s,s} f_s + \sum_{s=t}^{T} \int_{\Theta^s} \left(\frac{1}{R^s} \left[1 - \left(\frac{\tau_s^y(\theta)}{1-\tau_s^y(\theta)} \right) \frac{\gamma_s(\theta) - \xi_s(\theta)}{1 + \epsilon_s(\theta) - \gamma_s(\theta)} \right] \frac{U_{E,s}}{U_{c,s}} S_{T+1-s,s} \right)$$

Here, the optimal pollution tax differs from the Pigouvian level depending on the sign of $\gamma - \xi$. Although neither the structure of labor income taxation nor that of capital income taxation change from the presence of the externality, fiscal concerns still affect the optimal commodity tax. First, the planner distorts the choice of consumption goods to encourage the good that is most complementary with labor supply. Second, as in Jacobs and de Mooij (2015), the planner has a "virtual subsidy" on the provision of environmental quality. In this different formulation, the pollution tax component of the commodity wedge is higher than the Pigouvian level when $\gamma_s > \xi_s$. In words, if consumption is more complementary with labor than the environment, then the optimal pollution tax is lower than the Pigouvian level as distorting consumption choices discourages labor supply more than improving environmental quality encourages it. Unlike its static counterpart, the dynamic "equity-adjusted" pollution tax depends on the expectation of all possible future paths for the labor income tax which determine how strongly the planner needs to distort provision of environmental quality.

Since integrated assessment models from which quantitative values of a carbon tax are derived are dynamic, this extension of Jacobs and de Mooij 2015 to a dynamic environment can allow one to adjust the carbon tax to redistributive concerns, provided they can calibrate all the relevant parameters. Since quantifying optimal dynamic taxes is already a complex task without environnemental considerations, simplifying assumptions need to be made in practice. Veracierto (2023) finds that aggregate dynamics of a Mirrleesian economy do not differ from those of a neoclassical economy in the special case of logarithmic utility, so one possible path forward is assuming that preferences are logarithmic and assuming that income taxes are already set optimally such that one only needs to adjust the optimal carbon tax formula using existing labor income taxes and calibrated elasticities.

5 Constrained-optimal dynamic non-linear wedges with externalities

In practice, we may consider the current levels of labor, capital, or commodity taxation to be suboptimal. This amounts to solving the planner's problem with an added constraint that the wedge(s) must equal an exogenous schedule, where it can only optimize with respect to the remaining free wedge(s). I consider two possible cases: (i) the commodity wedge is fixed and (ii) the labor and savings wedges are fixed. In the first case, we would like to know how the labor and savings wedges would be optimally set if the commodity wedge were suboptimal. In the second case, we would like to know how the commodity wedge would be set if the other wedges were suboptimal. Note that I do not restrict the form of the exogenous schedule and it may indeed be non-linear. Hence, when I say that a wedge is underdistorted or overdistorted, I am talking about the wedge for individual θ.

5.1 Constrained-optimal commodity wedge (suboptimal labor wedge)

Suppose the labor wedge is constrained to exogenous schedule $\bar{\tau}^y(\theta)$. How does this change the optimal commodity wedge?

Proposition 5.1.1 (Third best commodity wedge (constrained labor wedge)).
The optimal commodity wedge when the labor wedge is constrained to $\bar{\tau}^y(\theta)$ is

$$\frac{\tau^g}{1+\tau^g} = \Delta + \left[\left(\frac{\bar{\tau}^y}{1-\bar{\tau}^y}\right) + \nu^y \Gamma_l^y\right] \frac{\gamma - \kappa}{1+\epsilon-\gamma} + \nu^y \Gamma_q^y$$

where

$$\begin{aligned}\Delta = &\int \left(-D'(S)\theta l - \left(1 - \left[\left(\frac{\bar{\tau}^y}{1-\bar{\tau}^y}\right) + \nu^y \Gamma_l^y\right] \frac{\gamma - \xi_s}{1+\epsilon-\gamma} + \nu^y \Gamma_S^y\right) \frac{U_S}{U_c}\right) f\ S_2 \\ &+ \frac{1}{R}\int \left(-D'(S')\theta' l' - \left(1 - \left[\left(\frac{\bar{\tau}^{y'}}{1-\bar{\tau}^{y'}}\right) + \nu^{y'} \Gamma_l^{y'}\right] \frac{\gamma' - \xi_s'}{1+\epsilon'-\gamma'} + \nu^{y'} \Gamma_S^{y'}\right) \frac{U_S'}{U_c'}\right) f\ S_1\end{aligned}$$

$$\begin{aligned}\Gamma_l^y &= \frac{U_{ll}}{\theta U_l f} - \frac{U_{cl}}{\theta U_c f} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c f} + \frac{U_{cl}}{U_l f}\right) \\ \Gamma_q^y &= \frac{U_{ql} - U_{cl}}{U_c \theta f} + (1-\bar{\tau}^y)\frac{U_{cq} - U_{cc}}{U_c f} \\ \Gamma_S^y &= \frac{U_{cl} - U_{lS}}{\theta U_c f} + (1-\bar{\tau}^y)\frac{U_{cq} - U_{cS}}{U_c f}\end{aligned}$$

Proof. See appendix C.1 □

This equation shows us how the planner optimally adjusts the commodity wedge when they cannot freely adjust the labor wedge. If $\nu^y(\theta) > 0$, this means the labor wedge is too low for the type θ individual. If $\nu^y(\theta) < 0$, this means the labor wedge is too high for the type θ individual.

First, note that we recover the second-best formula when $\nu^y(\theta) = 0$, meaning the labor wedge is set optimally. Second, how the planner optimally responds to a suboptimal labor wedge does not only depend on the sign of $\nu^y(\theta)$, but also on the structure of preferences. For example, let's set $\Delta = 0$ for the moment and focus on the overall commodity wedge.

$$\frac{\tau^g}{1+\tau^g} = \left[\left(\frac{\bar{\tau}^y}{1-\bar{\tau}^y}\right) + \nu^y \Gamma_l^y\right] \frac{\gamma - \kappa}{1 + \epsilon - \gamma} + \nu^y \Gamma_q^y$$

Suppose that preferences are fully separable, concave in consumption ($U_{cc} \leq 0$) and convex in labor supply ($U_{ll} \geq 0$). Let $\nu^y > 0$ (i.e. labor wedge too low for individual with skill history θ).

$$\frac{\tau^g}{1+\tau^g} = -\nu^y \left[\underbrace{(1-\bar{\tau}^y)}_{(+)} \underbrace{\frac{U_{cc}}{U_c f}}_{(-)}\right]$$

Under the aforementioned assumptions, the sign of the term associated with Γ_q^y is negative. Hence, if we take into account the negative, the commodity wedge becomes positive. That is, when the labor wedge is too low for a particular individual, the planner will compensate by taxing the good even if they usually would not want to do so in a second-best setting. If $\nu^y < 0$ and the labor wedge is too high for a particular individual, the sign of the effect will flip and the planner will subsidize the good to make up for overdistorting labor supply. If $\Delta > 0$,

$$\frac{\tau^g}{1+\tau^g} = \Delta - \nu^y \left[(1-\bar{\tau}^y)\frac{U_{cc}}{U_c f}\right]$$

If we stay in the case with fully separable utility, the environmental correction term is Pigouvian. Indeed, $\gamma = \xi_s$ such that the term associated with Γ_S^y drops out. Moreover, Γ_l^S also drops out. Hence, if the good generates pollution, underdistorting labor will mean that the planner should compensate by distorting the polluting good above the Pigouvian level. On the contrary, overdistorting labor implies a need to distort the good below the Pigouvian level.

5.2 Constrained-optimal commodity wedge (suboptimal savings wedge)

Suppose the savings wedge is constrained at $\bar{\tau}^s(\theta)$. Define $g(w, w_2, \theta, Q) \equiv \mathbb{E}_t[U_{c,t+1}]$. How does this change the optimal commodity wedge?

TBA

5.3 Constrained-optimal labor and savings wedges

Conversely, suppose the commodity wedge is constrained to $\bar{\tau}^g(\theta)$ and the labor and savings wedges are free. For example suppose the commodity wedge cannot be set optimally due to practical considerations or political feasibility.

Proposition 5.3.1 (Third-best labor wedge (constrained commodity wedge)). *The optimal labor wedge when the commodity wedge is constrained to $\bar{\tau}^g(\theta)$ is*

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \left[\left(\frac{\tau_{t-1}^y}{1-\tau_{t-1}^y}\right) + \nu_{t-1}^g \Gamma_{q,t-1}^g\right] D_t(\theta) + \nu_t^g \Gamma_{l,t}^g$$

where

$$\begin{aligned}
A_t(\theta) &\equiv (1+\epsilon_t(\theta)-\gamma_t(\theta)) \\
B_t(\theta) &\equiv \frac{1-F_t(\theta)}{\theta_t f_t(\theta)} \\
C_t(\theta) &\equiv \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right) d\tilde{x}\right] (1+\nu_t^g\Gamma_{q,t}^g - \lambda_{1,t}U_{c,t}(x))\frac{f_t(x)dx}{1-F_t(\theta)} \\
D_t(\theta) &\equiv \frac{A_t(\theta)}{A_{t-1}(\theta)}\frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)}\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\, dx}{\theta_t f_t(\theta)}
\end{aligned}$$

and

$$\begin{aligned}
\Gamma_{l,t}^g &\equiv \frac{U_{cq}}{U_c f} - \frac{U_{ql}}{U_l f} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c f} - \frac{U_{cq}}{U_l f}\right) \\
\Gamma_{q,t}^g &\equiv \frac{U_{cq}}{U_c f} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c f}
\end{aligned}$$

Proof. See appendix C.3. □

Analogously to the previous case, $\nu^g(\theta) > 0$ means the commodity wedge is too low for the type θ individual, while $\nu^g(\theta) < 0$ means the commodity wedge is too high for the type θ individual. Once again, note that we recover the second-best formula when the commodity wedge just so happens to be set optimally, such that $\nu^g(\theta) = 0$. Compared to the second-best formula, which was unchanged from the no-externalities formula, the intratemporal C term and the intertemporal D terms change. Moreover, a new term $\nu^g\Gamma_l^g$ appears. First, let's consider the case where preferences are fully separable and suppose $\nu^g > 0$.

In this case, the ABCD formula simplifies to

$$\frac{\tau_t^y}{1-\tau_t^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \left[\left(\frac{\tau_{t-1}^y}{1-\tau_{t-1}^y}\right) + \nu_{t-1}^g \Gamma_{q,t-1}^g\right] D_t(\theta) + \nu_t^g \Gamma_{l,t}^g$$

where

$$\begin{aligned}
A_t(\theta) &\equiv (1+\epsilon_t(\theta)) \\
B_t(\theta) &\equiv \frac{1-F_t(\theta)}{\theta_t f_t(\theta)} \\
C_t(\theta) &\equiv \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})}\right) d\tilde{x}\right] (1+\nu_t^g \Gamma_{q,t}^g - \lambda_{1,t} U_{c,t}(x)) \frac{f_t(x)dx}{1-F_t(\theta)} \\
D_t(\theta) &\equiv \frac{A_t(\theta)}{A_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1}\int_\theta^\infty f_{2,t}(x)\, dx}{\theta_t f_t(\theta)}
\end{aligned}$$

and

$$\begin{aligned}
\Gamma_{l,t}^g &\equiv -(1+\bar{\tau}^g)\frac{U_{cc}}{U_c f} \\
\Gamma_{q,t}^g &\equiv -(1+\bar{\tau}^g)\frac{U_{cc}}{U_c f}
\end{aligned}$$

If utility is increasing and concave in consumption, $\frac{U_{cc}}{U_c f} < 0$. Moreover, if we assume $\bar{\tau}^g \in (-1, 1)$ (i.e. it is at most a 100% tax or subsidy), then $1-\bar{\tau}^g \geq 0$. Thus, $\Gamma_l^g = \Gamma_{q,t}^g \geq 0$. *Ceteris paribus*, the C term then increases in comparison to the second-best. Moreover, the last term further increases the labor wedge. If we had $\nu^g < 0$ in the past period, then the D term also contributes to increasing the labor wedge. In sum, when the commodity wedge is too low for type θ, the planner optimally responds by increasing labor distortions for this person. In other words, the planner compensates for underdistorted goods choices by overdistorting labor supply.

In the case where preferences are fully separable, the commodity wedge should only be composed of a Pigouvian correction term. The interpretation of $\nu^g > 0$ then becomes that the polluting good is taxed under the social cost of carbon. Hence, the increase in labor distortions suggests that one way to compensate for under-taxing carbon might be to make the labor income tax more progressive. When preferences are non-separable, the optimal adjustment of the labor wedge is unclear and depends on the structure of preferences. In some cases, underdistorting the polluting good may require *underdistorting* labor supply as well.

Proposition 5.3.2 (Third-best savings wedge (constrained commodity wedge)). *The optimal savings wedge when the commodity wedge is constrained to $\bar{\tau}^g(\theta)$ is*

Proof. See appendix C.4 (TBA). □

6 Simulations

TBA

7 Conclusion

Using a dynamic Mirrlees framework, which features an explicit motive for income taxation, extended to feature consumption externalities, I find that the optimal pollution tax is only adjusted for inequality when there are non-separabilities in preferences between consumption, labor, and pollution. Otherwise, taxes are Pigouvian even in the presence of labor and capital income taxation, which shows that the dynamic Ramsey result that tax distortions justify deviating from Pigouvian taxation does not necessarily hold. Moreover, the optimal formulae for labor and savings wedges remain unchanged, suggesting that a properly designed commodity wedge is fully capable of internalizing the externality while respecting the social planner's redistributive goals.

When considering a third-best setting where one of the wedges is fixed at an arbitrary exogenous schedule, I find that the other wedges must be adjusted to compensate for the (potential) suboptimality of the fixed wedge. However, the direction of the adjustment depends on the structure of preferences. For example, when preferences are fully separable, underdistorting labor supply requires overdistorting commodity choices. If preferences are non-separable and some cross-elasticities are sufficiently large, the direction of the adjustment might flip.

There is much room for further study. First, this paper focuses on wedges rather than tax implementation, so an interesting avenue of research may be assessing how the optimal commodity wedge may be replicated in practice. Second, the paper features a single generation of potentially infinitely-lived individuals although climate change is in practice very much an intergenerational problem. Therefore, extending a heterogeneous framework to feature overlapping generations is another promising next step. Outside of the environmental context, since the optimal commodity wedge is a time-varying object in this dynamic model, one might also wonder if and how age-dependent commodity taxes may be used in a fiscal mix. Although age-dependent commodity taxation may at first glance appear a fanciful notion, positive age discrimination in the price of certain goods and services for young or old consumers or indeed discounts for low-income consumers already somewhat acts as implementing a time-varying income-based commodity wedge[3].

[3] Of course, whether this is done for social or economic reasons is another question. However, the point is that it *can* be done. Future work could evaluate the possible deliberate use of age-dependent income-dependent commodity wedges to further efficiency or equity objectives.

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A Optimal wedge derivation without externalities

A.1 Commodity wedge

Because $U(c, q, l)$ can be written as $U(c, q, y/\theta)$, I can write U_θ as $-U_l l/\theta$. The Hamiltonian to this problem is as follows.

$$
\begin{aligned}
\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta)\right] f \\
& + \psi \left[-U_l(c, q, l)\frac{l}{\theta} + \beta w_2\right] \\
& - \lambda_1 u(\theta) f_t \\
& + \lambda_2 u(\theta) f_2 \\
& + \phi \left[u - U(c, q, l) - \beta w\right]
\end{aligned}
$$

This yields the following first-order conditions.

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql}\frac{l}{\theta} = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l = \frac{1}{\theta}\psi \left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi = \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f = -\psi\beta
\end{aligned}
$$

Because the first-order conditions are unchanged, aside from the presence of $\frac{\partial \mathcal{H}}{\partial q}$ which is not involved in the derivation of non-commodity wedges, the steps to derive the expression for the optimal labor income wedge and the optimal savings wedge are identical to Golosov, Troshkin, et al. (2016). Hence, I refer to this paper for details.

Next, I use $\frac{\partial \mathcal{H}}{\partial c}$ to write $\phi = \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c}$, which I substitute in the first-order conditions.

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql} \frac{l}{\theta} = \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} U_l = \frac{1}{\theta} \psi \left[U_{ll} l + U_l \right] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 \bar{\alpha}_t f + \lambda_2 f_2 + \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} = \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta
\end{aligned}
$$

From here, I rewrite $\frac{\partial \mathcal{H}}{\partial q}$ using $1 + \tau^g = \frac{U_q}{U_c}$.

$$
1 + \tau^g = \frac{f - \psi U_{ql} l \frac{1}{\theta}}{f - \psi U_{cl} l \frac{1}{\theta}}
$$

I then rewrite the equation in terms of elasticities using $\gamma_t(\theta) \equiv \frac{U_{cl,t}(\theta) l_t(\theta)}{U_{c,t}(\theta)}$ and $\kappa_t(\theta) \equiv \frac{U_{ql,t}(\theta) l_t(\theta)}{U_{c,t}(\theta)}$, substitute $\frac{\psi}{\theta f}$, and rearrange terms.

$$
1 + \tau^g = \frac{1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\kappa (1 + \tau^g)}{1 + \epsilon - \gamma}}{1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma}{1 + \epsilon - \gamma}}
$$

If we isolate $1 + \tau^g$,

$$
(1 + \tau^g) \left[1 + \frac{\left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\kappa}{1 + \epsilon - \gamma}}{1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma}{1 + \epsilon - \gamma}} \right] = \frac{1}{1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma}{1 + \epsilon - \gamma}}
$$

After simplification,

$$
(1 + \tau^g) \left[\frac{1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma - \kappa}{1 + \epsilon - \gamma}}{1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma}{1 + \epsilon - \gamma}} \right] = \frac{1}{1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma}{1 + \epsilon - \gamma}}
$$

Hence,

$$1+\tau^g = \frac{1}{1-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\gamma-\kappa}{1+\epsilon-\gamma}}$$

Note that

$$\tau^g = \frac{\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\gamma-\kappa}{1+\epsilon-\gamma}}{1-\left(\frac{\tau^y}{1-\tau^y}\right)\frac{\gamma-\kappa}{1+\epsilon-\gamma}}$$

If I write this in $\frac{\tau^g}{1+\tau^g}$ form, I get the desired result.

$$\frac{\tau^g}{1+\tau^g} = \left(\frac{\tau^y}{1-\tau^y}\right)\frac{\gamma-\kappa}{1+\epsilon-\gamma}$$

B Optimal wedge derivation with externalities

B.1 Commodity wedge

Because we can write $U(c,q,l,S)$ as $U(c,q,\frac{y}{D(S)\theta},S)$, $U_\theta = -U_l\frac{l}{\theta}$. After substituting U_θ in the planner's problem, its Hamiltonian becomes

$$
\begin{aligned}
\mathcal{H} = &\left[c + q - D(S)\theta l + \frac{1}{R}V_{t+1}\left(w, w_2, \theta, \int q(\theta)f(\theta)d\theta\right)\right] f \\
&+ \psi\left[-U_l(c,q,l,S)\frac{l}{\theta} + \beta w_2\right] \\
&- \lambda_1 u(\theta) f_t \\
&+ \lambda_2 u(\theta) f_2 \\
&+ \phi\left[u - U(c,q,l,S) - \beta w\right] \\
&- \mu\left[S - S\left(\cdots, \int q(\theta)\ f(\theta)d\theta\right)\right]
\end{aligned}
$$

The first-order conditions are as follows.

$$
\begin{aligned}
\frac{\partial\mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} &&= \phi U_c \\
\frac{\partial\mathcal{H}}{\partial q} &: f + \frac{1}{R}\frac{\partial V}{\partial Q}f(\theta) - \psi U_{ql}\frac{l}{\theta} + \mu S_2 f &&= \phi U_q \\
\frac{\partial\mathcal{H}}{\partial l} &: -D(S)\theta f - \phi U_l &&= \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\frac{\partial\mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &&= \dot{\psi} \\
\frac{\partial\mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w}f &&= \phi\beta \\
\frac{\partial\mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2}f &&= -\psi\beta \\
\frac{\partial\mathcal{H}}{\partial S} &: \int\left(-D'(S)\theta l f - \psi U_{lS}\frac{l}{\theta} - \phi U_S\right) &&= \mu
\end{aligned}
$$

First, let's start with deriving the labor wedge. We can substitute ϕ using the FOC w.r.t. c.

$$\phi = \frac{f - \psi U_{cl}\frac{l}{\theta}}{U_c}$$

This yields the following modified FOCs.

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R}\frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql}\frac{l}{\theta} + \mu S_2 f && = \frac{f - \psi U_{cl}\frac{l}{\theta}}{U_c} U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -D(S)\theta f - \frac{f - \psi U_{cl}\frac{l}{\theta}}{U_c} U_l && = \frac{1}{\theta}\psi\left[U_{ll} l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \frac{f - \psi U_{cl}\frac{l}{\theta}}{U_c} && = \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w} f && = \frac{f - \psi U_{cl}\frac{l}{\theta}}{U_c}\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2} f && = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int \left(-D'(S)\theta l f - \psi U_{lS}\frac{l}{\theta} - \frac{f - \psi U_{cl}\frac{l}{\theta}}{U_c} U_S\right) && = \mu
\end{aligned}$$

Using $\epsilon_t(\theta) \equiv \frac{U_{ll,t}(\theta) l_t(\theta)}{U_{l,t}(\theta)}$ and $\gamma_t(\theta) \equiv \frac{U_{cl,t}(\theta) l_t(\theta)}{U_{c,t}(\theta)}$, we can rewrite $\frac{\partial \mathcal{H}}{\partial l}$.

$$\begin{aligned}
-D(S)\theta f - \frac{f - \psi U_{cl}\frac{l}{\theta}}{U_c} U_l = \frac{1}{D(S)\theta}\psi\left[U_{ll} l + U_l\right] &\implies -D(S)\theta f - \frac{U_l}{U_c} f + \psi \frac{U_{cl}}{U_c}\frac{U_l l}{\theta} = -\frac{1}{\theta}\psi\left[\frac{U_{ll} l + U_l}{U_l}\right](-U_l) \\
&\implies -\left(\frac{U_l}{D(S)\theta U_c} + 1\right) D(S)\theta f = -\frac{1}{\theta}\psi\left[\frac{U_{ll} l + U_l}{U_l}\right](-U_l) - \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} \\
&\implies -\left(\frac{U_l}{D(S)\theta U_c} + 1\right) D(S)\theta f = -\frac{1}{\theta}\psi(\epsilon + 1)(-U_l) - \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} \\
&\implies \left(\frac{U_l}{D(S)\theta U_c} + 1\right) D(S)\theta f = \frac{1}{\theta}\psi\left(\epsilon + 1 - \frac{U_{cl} l}{U_c}\right)(-U_l) \\
&\implies \left(\frac{U_l}{D(S)\theta U_c} + 1\right) D(S)\theta f = \frac{1}{\theta}\psi(1 + \epsilon - \gamma)(-U_l)
\end{aligned}$$

If there are damages to output, $\tau^y = 1 + \frac{U_l}{D(S)\theta U_c}$

$$
\begin{aligned}
\tau^y D(S)\theta f = \frac{1}{\theta}\psi(1+\epsilon-\gamma)(-U_l) &\implies \tau^y = \frac{\psi}{D(S)\theta^2 f}(1+\epsilon-\gamma)(-U_l) \\
&\implies \tau^y = \frac{-\psi U_l}{D(S)\theta^2 f}(1+\epsilon-\gamma) \\
&\implies \tau^y = \frac{-\psi U_y}{\theta f}(1+\epsilon-\gamma) \\
&\implies \tau^y = -U_c\frac{\psi U_y}{U_c\theta f}(1+\epsilon-\gamma) \\
&\implies \tau^y = U_c\frac{\psi(1-\tau^y)}{\theta f}(1+\epsilon-\gamma) \\
&\implies \frac{\tau^y}{1-\tau^y} = \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma)
\end{aligned}
$$

From here, the steps are the same as in Golosov, Troshkin, et al. (2016), meaning the labor wedge is unchanged. Likewise, because the savings wedge involves $\frac{\partial\mathcal{H}}{\partial u}$ and $\frac{\partial\mathcal{H}}{\partial w}$ which are unchanged, the savings wedge is also unchanged.

Now onto the commodity wedge. First, recall the first-order conditions.

$$
\begin{aligned}
\frac{\partial\mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} &&= \phi U_c \\
\frac{\partial\mathcal{H}}{\partial q} &: f + \frac{1}{R}\frac{\partial V}{\partial Q}f(\theta) - \psi U_{ql}\frac{l}{\theta} + \mu S_2 f &&= \phi U_q \\
\frac{\partial\mathcal{H}}{\partial l} &: -D(S)\theta f - \phi U_l &&= \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\frac{\partial\mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &&= \dot{\psi} \\
\frac{\partial\mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w}f &&= \phi\beta \\
\frac{\partial\mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2}f &&= -\psi\beta \\
\frac{\partial\mathcal{H}}{\partial S} &: \int\left(-D'(S)\theta l f - \psi U_{lS}\frac{l}{\theta} - \phi U_S\right) &&= \mu
\end{aligned}
$$

$\frac{\partial V_t}{\partial Q_t}$ can be found using the envelope condition.

$$\frac{\partial \mathcal{H}}{\partial Q_{t-1}} = \mu_t S_{1,t} \implies \frac{\partial V_t}{\partial Q_t} = \mu_{t+1} S_{1,t+1}$$

After replacing μ and $\frac{\partial V}{\partial Q}$,

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} && = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R} \int \left(-D'(S')\theta' l' f' - \psi U'_{lS} \frac{l'}{\theta'} - \phi U'_S \right) S_1 f - \psi U_{ql} \frac{l}{\theta} + \int \left(-D'(S)\theta l f - \psi U_{lS} \frac{l}{\theta} - \phi U_S \right) S_2 f && = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -D(S)\theta f - \phi U_l && = \frac{1}{\theta} \psi \left[U_{ll} l + U_l \right] \\
\frac{\partial \mathcal{H}}{\partial u} &: -D(S)\lambda_1 \bar{\rho}_t f + \lambda_2 f_2 + \phi && = \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f && = \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f && = -\psi \beta
\end{aligned}$$

Then, I use the first equation to substitute ϕ.

$$\phi = \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c}$$

$$\frac{\partial \mathcal{H}}{\partial q} : f + \frac{1}{R} \int \left(-D'(S')\theta' l' f' - \psi' U'_{lS} \frac{l'}{\theta'} - \frac{f' - \psi' U'_{cl} \frac{l'}{\theta'}}{U'_c} U'_S \right) S_1 f - \psi U_{ql} \frac{l}{\theta} + \int \left(-D'(S)\theta l f - \psi U_{lS} \frac{l}{\theta} - \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} U_S \right) S_2 f = \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} U_c$$

$$\frac{\partial \mathcal{H}}{\partial l} : -D(S)\theta f - \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} U_l = \frac{1}{\theta} \psi \left[U_{ll} l + U_l \right.$$

$$\frac{\partial \mathcal{H}}{\partial u} : -\lambda_1 f + \lambda_2 f_2 + \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} = \dot{\psi}$$

$$\frac{\partial \mathcal{H}}{\partial w} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \frac{f - \psi U_{cl} \frac{l}{\theta}}{U_c} \beta$$

$$\frac{\partial \mathcal{H}}{\partial w_2} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta$$

Now, I rewrite $\frac{\partial \mathcal{H}}{\partial q}$ using $1 + \tau^g = \frac{U_q}{U_c}$. I also define $\xi \equiv \frac{U_{lS} l}{U_S}$ as the elasticity between pollution damages and labor supply. I express the equation such that it resembles last section's commodity tax formula, with the commodity taxation term isolated on the right-hand side.

$$\frac{\tau_t^g(\theta) - \Delta_t}{1 + \tau_t^g(\theta)} = \left(\frac{\tau_t^y(\theta)}{1 - \tau_t^y(\theta)} \right) \frac{\gamma_t(\theta) - \kappa_t(\theta)}{1 + \epsilon_t(\theta) - \gamma_t(\theta)}$$

$$\Delta_t \equiv \sum_{s=t}^{T} \int_{\Theta^s} \frac{1}{R^s} \left[D'(S)\theta l \right] S_{T+1-s,s} f_s + \sum_{s=t}^{T} \int_{\Theta^s} \frac{1}{R^s} \left[1 - \left(\frac{\tau_s^y(\theta)}{1 - \tau_s^y(\theta)} \right) \frac{\gamma_s(\theta) - \xi_s(\theta)}{1 + \epsilon_s(\theta) - \gamma_s(\theta)} \right] \frac{U_{S,s}}{U_{c,s}} S_{T+1-s,s} \, f_s$$

C Constrained-optimal wedge derivations

C.1 Constrained-optimal commodity wedge (suboptimal labor wedge)

$$V_t(\hat{w}, \hat{w}_2, \theta_-, Q_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - D(S)y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, Q) \right] f_t(\theta|\theta_-) d\theta$$

s.t.

$$\begin{aligned}
\dot{u}(\theta, S) &= U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right) + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta) f_t(\theta|\theta_-) d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta \\
u(\theta, S) &= U \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, E \right) + \beta w(\theta) \\
S &= S \left(\cdots, \int q(\theta) f(\theta) d\theta \right) \\
1 - \bar{\tau}^y(\theta) &= - \frac{U_l \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right)}{\theta U_c \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right)}
\end{aligned}$$

This yields this Hamiltonian, where ν^y is the Lagrange multiplier on the labor wedge constraint.

$$\begin{aligned}
\mathcal{H} = & \left[c + q - D(S)\theta l + \frac{1}{R} V_{t+1} \left(w, w_2, \theta, \int q(\theta) f(\theta) d\theta \right) \right] f \\
& + \psi \left[-U_l(c, q, l, S) \frac{l}{\theta} + \beta w_2 \right] \\
& - \lambda_1 u(\theta) f_t \\
& + \lambda_2 u(\theta) f_2 \\
& + \phi \left[u - U(c, q, l, S) - \beta w \right] \\
& - \mu \left[S - S \left(\cdots, \int q(\theta)\ f(\theta) d\theta \right) \right] \\
& + \nu^y \left[1 - \bar{\tau}^y(\theta) + \frac{U_l\left(c, q, l, S\right)}{\theta U_c\left(c, q, l, S\right)} \right]
\end{aligned}$$

The first-order conditions are as follows.

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} + \nu^y\left(\frac{U_{cl}}{\theta U_c} - \frac{U_l U_{cc}}{\theta U_c^2}\right) &&= \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R}\frac{\partial V}{\partial Q}f(\theta) - \psi U_{ql}\frac{l}{\theta} + \mu S_2 f + \nu^y\left(\frac{U_{ql}}{\theta U_c} - \frac{U_l U_{cq}}{\theta U_c^2}\right) &&= \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -D(S)\theta f - \phi U_l + \nu^y\left(\frac{U_{ll}}{\theta U_c} - \frac{U_l U_{cl}}{\theta U_c^2}\right) &&= \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &&= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w}f &&= \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2}f &&= -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int\left(-D'(S)\theta l f - \psi U_{lS}\frac{l}{\theta} - \phi U_S + \nu^y\left(\frac{U_{lS}}{\theta U_c} - \frac{U_l U_{cS}}{\theta U_c^2}\right)\right) &&= \mu
\end{aligned}
$$

Using the labor wedge constraint $1 - \bar{\tau}^y(\theta) = -\frac{U_l}{\theta U_c}$,

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} + \nu^y\left(\frac{U_{cl}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right) &&= \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R}\frac{\partial V}{\partial Q}f(\theta) - \psi U_{ql}\frac{l}{\theta} + \mu S_2 f + \nu^y\left(\frac{U_{ql}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cq}}{U_c}\right) &&= \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -D(S)\theta f - \phi U_l + \nu^y\left(\frac{U_{ll}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cl}}{U_c}\right) &&= \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &&= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w}f &&= \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2}f &&= -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int\left(-D'(S)\theta l f - \psi U_{lS}\frac{l}{\theta} - \phi U_S + \nu^y\left(\frac{U_{lS}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cS}}{U_c}\right)\right) &&= \mu
\end{aligned}
$$

We can substitute ϕ in $\frac{\partial \mathcal{H}}{\partial q}$ using the FOC w.r.t. c and then rewrite it using $1+\tau^g = \frac{U_q}{U_c}$.

$$\phi = \frac{f - \psi U_{cl}\frac{l}{\theta} + \nu^y\left(\frac{U_{cl}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)}{U_c}$$

$$1+\tau^g = \frac{f + \frac{1}{R}\frac{\partial V}{\partial Q}f - \psi U_{ql}\frac{l}{\theta} + \mu S_2 f + \nu^y\left(\frac{U_{ql}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cq}}{U_c}\right)}{f - \psi U_{cl}\frac{l}{\theta} + \nu^y\left(\frac{U_{cl}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)}$$

Next, I find the expression for ψ by rewriting $\frac{\partial \mathcal{H}}{\partial l}$.

$$\begin{aligned}
& -D(S)\theta f - \phi U_l + \nu^y\left(\frac{U_{ll}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cl}}{U_c}\right) = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\implies & -D(S) - \frac{f - \psi U_{cl}\frac{l}{\theta} + \nu^y\left(\frac{U_{cl}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)}{U_c}U_l + \nu^y\left(\frac{U_{ll}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cl}}{U_c}\right) = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\implies & -D(S)\theta f - \frac{U_l}{U_c}f + \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} + \nu^y\left(-\frac{U_{cl}}{\theta U_c} - (1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)\frac{U_l}{U_c} + \nu^y\left(\frac{U_{ll}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cl}}{U_c}\right) = -\frac{1}{\theta}\psi\left[\frac{U_{ll}l + U_l}{U_l}\right](-U_l) \\
\implies & -D(S)\left(1 + \frac{U_l}{D(S)\theta U_c}\right)\theta f + \nu^y\left(-\frac{U_{cl}}{\theta U_c} - (1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)\frac{U_l}{U_c} + \nu^y\left(\frac{U_{ll}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cl}}{U_c}\right) = -\frac{1}{\theta}\psi(\epsilon+1)(-U_l) - \psi\frac{U_{cl}}{U_c}\frac{U}{} \\
\implies & -D(S)\left(1 + \frac{U_l}{D(S)\theta U_c}\right)\theta f + \nu^y\left(-\frac{U_{cl}}{\theta U_c} - (1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)\frac{U_l}{U_c} + \nu^y\left(\frac{U_{ll}}{\theta U_l} + (1-\bar{\tau}^y)\frac{U_{cl}}{U_l}\right)\frac{U_l}{U_c} = -\frac{1}{\theta}\psi(1+\epsilon-\gamma)(-U_l) \\
\implies & -D(S)\left(1 + \frac{U_l}{D(S)\theta U_c}\right)\theta f + \nu^y\left(\frac{U_{ll}}{\theta U_l} - \frac{U_{cl}}{\theta U_c} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c} + \frac{U_{cl}}{U_l}\right)\right)\frac{U_l}{U_c} = -\frac{1}{\theta}\psi(1+\epsilon-\gamma)(-U_l)
\end{aligned}$$

Because $\bar{\tau}^y = 1 + \frac{U_l}{D(S)\theta U_c}$,

$$
\begin{aligned}
& -D(S)\bar{\tau}^y\theta f + \nu^y\left(\frac{U_{ll}}{\theta U_l} - \frac{U_{cl}}{\theta U_c} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c} + \frac{U_{cl}}{U_l}\right)\right)\frac{U_l}{U_c} = -\frac{1}{\theta}\psi(1+\epsilon-\gamma)(-U_l) \\
\implies & \bar{\tau}^y - \nu^y\left(\frac{U_{ll}}{\theta U_l} - \frac{U_{cl}}{\theta U_c} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c} + \frac{U_{cl}}{U_l}\right)\right)\frac{U_l}{D(S)\theta U_c f} = \frac{\psi}{\theta^2 f}(1+\epsilon-\gamma)(-U_l) \\
\implies & \bar{\tau}^y - \nu^y\left(\frac{U_{ll}}{\theta U_l} - \frac{U_{cl}}{\theta U_c} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c} + \frac{U_{cl}}{U_l}\right)\right)\frac{U_l}{D(S)\theta U_c f} = U_c\frac{\psi(1-\bar{\tau}^y)}{\theta f}(1+\epsilon-\gamma) \\
\implies & \bar{\tau}^y + \nu^y\left(\frac{U_{ll}}{\theta U_l f} - \frac{U_{cl}}{\theta U_c f} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c f} + \frac{U_{cl}}{U_l f}\right)\right)(1-\bar{\tau}^y) = U_c\frac{\psi(1-\bar{\tau}^y)}{\theta f}(1+\epsilon-\gamma) \\
\implies & \frac{\bar{\tau}^y}{1-\bar{\tau}^y} = \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma) - \nu^y\left(\frac{U_{ll}}{\theta U_l f} - \frac{U_{cl}}{\theta U_c f} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c f} + \frac{U_{cl}}{U_l f}\right)\right) \\
\implies & \frac{\psi U_c}{\theta f}(1+\epsilon-\gamma) = \frac{\bar{\tau}^y}{1-\bar{\tau}^y} + \nu^y\left(\frac{U_{ll}}{\theta U_l f} - \frac{U_{cl}}{\theta U_c f} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c f} + \frac{U_{cl}}{U_l f}\right)\right) \\
\implies & \psi = \left[\left(\frac{\bar{\tau}^y}{1-\bar{\tau}^y}\right) + \nu^y\left(\frac{U_{ll}}{\theta U_l f} - \frac{U_{cl}}{\theta U_c f} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c f} + \frac{U_{cl}}{U_l f}\right)\right)\right]\frac{\theta f}{U_c}\frac{1}{1+\epsilon-\gamma}
\end{aligned}
$$

Now we can substitute ψ back into the expression for τ^g and rewrite it in elasticity terms.

$$
\begin{aligned}
1+\tau^g &= \frac{f + \frac{1}{R}\frac{\partial V}{\partial Q}f - [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{ql}}{U_c}lf + \mu S_2 f + \nu^y\left(\frac{U_{ql}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cq}}{U_c}\right)}{f - [\cdots]\frac{\gamma}{1+\epsilon-\gamma}f + \nu^y\left(\frac{U_{cl}}{\theta U_c} + (1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)} \\
&= \frac{1 + \frac{1}{R}\frac{\partial V}{\partial Q} - [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{ql}}{U_q}l\frac{U_q}{U_c} + \mu S_2 + \nu^y\left(\frac{U_{ql}}{U_c\theta f} + (1-\bar{\tau}^y)\frac{U_{cq}}{U_c f}\right)}{1 - [\cdots]\frac{\gamma}{1+\epsilon-\gamma} + \nu^y\left(\frac{U_{cl}}{U_c\theta f} + (1-\bar{\tau}^y)\frac{U_{cc}}{U_c f}\right)} \\
&= \frac{1 + \frac{1}{R}\frac{\partial V}{\partial Q} - [\cdots]\frac{1}{1+\epsilon-\gamma}\kappa(1+\tau^g) + \mu S_2 + \nu^y\left(\frac{U_{ql}}{U_c\theta f} + (1-\bar{\tau}^y)\frac{U_{cq}}{U_c f}\right)}{1 - [\cdots]\frac{\gamma}{1+\epsilon-\gamma} + \nu^y\left(\frac{U_{cl}}{U_c\theta f} + (1-\bar{\tau}^y)\frac{U_{cc}}{U_c f}\right)}
\end{aligned}
$$

If we isolate $1+\tau^g$,

$$\frac{1+\frac{1}{R}\frac{\partial V}{\partial Q}+\mu S_2+\nu^y\left(\frac{U_{ql}}{U_c\theta f}+(1-\bar{\tau}^y)\frac{U_{cq}}{U_c f}\right)}{1-[\cdots]\frac{\gamma}{1+\epsilon-\gamma}+\nu^y\left(\frac{U_{cl}}{U_c\theta f}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c f}\right)}=(1+\tau^g)\left[1+\frac{[\cdots]\frac{\kappa}{1+\epsilon-\gamma}}{1-[\cdots]\frac{\gamma}{1+\epsilon-\gamma}+\nu^y\left(\frac{U_{cl}}{U_c\theta f}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c f}\right)}\right]$$
$$=(1+\tau^g)\left[\frac{1-[\cdots]\frac{\gamma-\kappa}{1+\epsilon-\gamma}+\nu^y\left(\frac{U_{cl}}{U_c\theta f}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c f}\right)}{1-[\cdots]\frac{\gamma}{1+\epsilon-\gamma}+\nu^y\left(\frac{U_{cl}}{U_c\theta f}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c f}\right)}\right]$$

Hence,

$$1+\tau^g=\frac{1+\frac{1}{R}\frac{\partial V}{\partial Q}+\mu S_2+\nu^y\left(\frac{U_{ql}}{U_c\theta f}+(1-\bar{\tau}^y)\frac{U_{cq}}{U_c f}\right)}{1-[\cdots]\frac{\gamma-\kappa}{1+\epsilon-\gamma}+\nu^y\left(\frac{U_{cl}}{U_c\theta f}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c f}\right)}$$

If I write this in $\frac{\tau^g}{1+\tau^g}$ form with $\Delta\equiv\mu S_2+\frac{1}{R}\frac{\partial V}{\partial Q}\frac{1}{R}\frac{\partial V}{\partial Q}$,

$$\frac{\tau^g}{1+\tau^g}=\Delta+[\cdots]\frac{\gamma-\kappa}{1+\epsilon-\gamma}+\nu^y\left(\frac{U_{ql}-U_{cl}}{U_c\theta f}+(1-\bar{\tau}^y)\frac{U_{cq}-U_{cc}}{U_c f}\right)$$

Now on to Δ. We can find $\frac{\partial V}{\partial Q}$ by the envelope condition as the future μ multiplied by S_1.

$$\begin{aligned}
\Delta=&\int\left(-D'(S)\theta lf-\psi U_{lS}\frac{l}{\theta}-\phi U_S+\nu^y\left(\frac{U_{lS}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cS}}{U_c}\right)\right)S_2\\
&+\frac{1}{R}\left(-D'(S)\theta lf-\psi U_{lS}\frac{l}{\theta}-\phi U_S+\nu^y\left(\frac{U_{lS}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cS}}{U_c}\right)\right)S_1\\
=&\int\left(-D'(S)\theta lf-\psi U_{lS}\frac{l}{\theta}-\frac{f-\psi U_{cl}\frac{l}{\theta}+\nu^y\left(\frac{U_{cl}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)}{U_c}U_S+\nu^y\left(\frac{U_{lS}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cS}}{U_c}\right)\right)S_2\\
&+\frac{1}{R}\int\left(-D'(S)\theta lf-\psi U_{lS}\frac{l}{\theta}-\frac{f-\psi U_{cl}\frac{l}{\theta}+\nu^y\left(\frac{U_{cl}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)}{U_c}U_S+\nu^y\left(\frac{U_{lS}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cS}}{U_c}\right)\right)S_1\\
=&\int\left(-D'(S)\theta lf-\psi U_{lS}\frac{l}{\theta}-\frac{U_S}{U_c}f+\psi\frac{U_{cl}}{U_c}\frac{l}{\theta}U_S-\frac{\nu^y\left(\frac{U_{cl}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)}{U_c}U_S+\nu^y\left(\frac{U_{lS}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cS}}{U_c}\right)\right)S_2\\
&+\frac{1}{R}\int\left(-D'(S)\theta lf-\psi U_{lS}\frac{l}{\theta}-\frac{U_S}{U_c}f+\psi\frac{U_{cl}}{U_c}\frac{l}{\theta}U_S-\frac{\nu^y\left(\frac{U_{cl}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cc}}{U_c}\right)}{U_c}U_S+\nu^y\left(\frac{U_{lS}}{\theta U_c}+(1-\bar{\tau}^y)\frac{U_{cS}}{U_c}\right)\right)S_1
\end{aligned}$$

$$
\begin{aligned}
\Delta = &\int \left(-D'(S)\theta l f - [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{lS}}{U_c} l f - \frac{U_S}{U_c} f + [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{cl}}{U_c} l \frac{U_S}{U_c} f + \cdots \right) S_2 \\
&+ \frac{1}{R}\int \left(-D'(S)\theta l f - [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{lS}}{U_c} l f - \frac{U_S}{U_c} f + [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{cl}}{U_c} l \frac{U_S}{U_c} f + \cdots \right) S_1 \\
\Delta = &\int \left(-D'(S)\theta l - [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{lS}}{U_S} l \frac{U_S}{U_c} - \frac{U_S}{U_c} + [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{cl}}{U_c} l \frac{U_S}{U_c} + \cdots \right) f\ S_2 \\
&+ \frac{1}{R}\int \left(-D'(S)\theta l - [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{lS}}{U_S} l \frac{U_S}{U_c} - \frac{U_S}{U_c} + [\cdots]\frac{1}{1+\epsilon-\gamma}\frac{U_{cl}}{U_c} l \frac{U_S}{U_c} + \cdots \right) f\ S_1 \\
\Delta = &\int \left(-D'(S)\theta l - \left([\cdots]\frac{\xi_{lS}}{1+\epsilon-\gamma} + 1 - [\cdots]\frac{\gamma}{1+\epsilon-\gamma} - \cdots \right) \frac{U_S}{U_c} \right) f\ S_2 \\
&+ \frac{1}{R}\int \left(-D'(S)\theta l - \left([\cdots]\frac{\xi_{lS}}{1+\epsilon-\gamma} + 1 - [\cdots]\frac{\gamma}{1+\epsilon-\gamma} - \cdots \right) \frac{U_S}{U_c} \right) f\ S_1 \\
\Delta = &\int \left(-D'(S)\theta l - \left(1 - [\cdots]\frac{\gamma-\xi_s}{1+\epsilon-\gamma} + \nu^y \left(\frac{U_{cl}-U_{lS}}{\theta U_c f} + (1-\bar{\tau}^y)\frac{U_{cq}-U_{cS}}{U_c f} \right) \right) \frac{U_S}{U_c} \right) f\ S_2 \\
&+ \frac{1}{R}\int \left(-D'(S)\theta l - \left(1 - [\cdots]\frac{\gamma-\xi_s}{1+\epsilon-\gamma} + \nu^y \left(\frac{U_{cl}-U_{lS}}{\theta U_c f} + (1-\bar{\tau}^y)\frac{U_{cq}-U_{cS}}{U_c f} \right) \right) \frac{U_S}{U_c} \right) f\ S_1
\end{aligned}
$$

To keeps things short,

$$
\begin{aligned}
\Gamma_l^y &= \frac{U_{ll}}{\theta U_l f} - \frac{U_{cl}}{\theta U_c f} - (1-\bar{\tau}^y)\left(\frac{U_{cc}}{U_c f} + \frac{U_{cl}}{U_l f} \right) \\
\Gamma_q^y &= \frac{U_{ql}-U_{cl}}{U_c \theta f} + (1-\bar{\tau}^y)\frac{U_{cq}-U_{cc}}{U_c f} \\
\Gamma_S^y &= \frac{U_{cl}-U_{lS}}{\theta U_c f} + (1-\bar{\tau}^y)\frac{U_{cq}-U_{cS}}{U_c f}
\end{aligned}
$$

Hence,

$$
\frac{\tau^g}{1+\tau^g} = \Delta + \left[\left(\frac{\bar{\tau}^y}{1-\bar{\tau}^y} \right) + \nu^y \Gamma_l^y \right] \frac{\gamma-\kappa}{1+\epsilon-\gamma} + \nu^y \Gamma_q^y
$$

where

$$
\begin{aligned}
\Delta = &\int \left(-D'(S)\theta l - \left(1 - \left[\left(\frac{\bar{\tau}^y}{1-\bar{\tau}^y} \right) + \nu^y \Gamma_l^y \right] \frac{\gamma-\xi_s}{1+\epsilon-\gamma} + \nu^y \Gamma_S^y \right) \frac{U_S}{U_c} \right) f\ S_2 \\
&+ \frac{1}{R}\int \left(-D'(S')\theta' l' - \left(1 - \left[\left(\frac{\bar{\tau}^{y'}}{1-\bar{\tau}^{y'}} \right) + \nu^{y'} \Gamma_l^{y'} \right] \frac{\gamma'-\xi_s'}{1+\epsilon'-\gamma'} + \nu^{y'} \Gamma_S^{y'} \right) \frac{U_S'}{U_c'} \right) f\ S_1
\end{aligned}
$$

C.2 Constrained-optimal commodity wedge (suboptimal savings wedge)

$$V_t(\hat{w}, \hat{w}_2, \theta_-, Q_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - D(S)y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, Q) \right] f_t(\theta|\theta_-) d\theta$$

$$\text{s.t.}$$

$$\dot{u}(\theta, S) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta) f_t(\theta|\theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta$$

$$u(\theta, S) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right) + \beta w(\theta)$$

$$S = S \left(\cdots, \int q(\theta) f(\theta) d\theta \right)$$

$$1 - \bar{\tau}^s(\theta) = \frac{1}{\beta R} \frac{U_c\left(c(\theta), q(\theta), l(\theta), S\right)}{g(w(\theta), w_2(\theta), \theta, Q)}$$

C.3 Constrained-optimal labor wedge (suboptimal commodity wedge)

$$V_t(\hat{w}, \hat{w}_2, \theta_-, Q_-) = \min_{c,q,l,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - D(S)\theta l(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, Q) \right] f_t(\theta|\theta_-) d\theta$$

s.t.

$$\begin{aligned}
\dot{u}(\theta, S) &= -U_l\left(c(\theta), q(\theta), l(\theta), S\right) \frac{1}{\theta} + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta) f_t(\theta|\theta_-) d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta \\
u(\theta, S) &= U\left(c(\theta), q(\theta), l(\theta), S\right) + \beta w(\theta) \\
S &= S\left(\cdots, \int q(\theta) f(\theta) d\theta\right) \\
1 + \bar{\tau}^g(\theta) &= \frac{U_q\left(c(\theta), q(\theta), l(\theta), S\right)}{U_c\left(c(\theta), q(\theta), l(\theta), S\right)}
\end{aligned}$$

We get this Hamiltonian, where ν^g is the Lagrange multiplier on the commodity wedge constraint.

$$\begin{aligned}
\mathcal{H} = &\left[c + q - D(S)\theta l + \frac{1}{R} V_{t+1}\left(w, w_2, \theta, \int q(\theta) f(\theta) d\theta\right) \right] f \\
&+ \psi \left[-U_l(c, q, l, S) \frac{l}{\theta} + \beta w_2 \right] \\
&- \lambda_1 u(\theta) f_t \\
&+ \lambda_2 u(\theta) f_2 \\
&+ \phi \left[u - U(c, q, l, S) - \beta w \right] \\
&- \mu \left[S - S\left(\cdots, \int q(\theta)\ f(\theta) d\theta\right) \right] \\
&+ \nu^g \left[1 + \bar{\tau}^g(\theta) + \frac{U_q\left(c, q, l, S\right)}{U_c\left(c, q, l, S\right)} \right]
\end{aligned}$$

The first-order conditions are as follows.

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} + \nu^g\left(\frac{U_{cq}}{U_c} - \frac{U_q U_{cc}}{U_c^2}\right) && = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R}\frac{\partial V}{\partial Q}f(\theta) - \psi U_{ql}\frac{l}{\theta} + \mu S_2 f + \nu^g\left(\frac{U_{qq}}{U_c} - \frac{U_q U_{cq}}{U_c^2}\right) && = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -D(S)\theta f - \phi U_l + \nu^g\left(\frac{U_{ql}}{U_c} - \frac{U_q U_{cq}}{U_c^2}\right) && = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi && = \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w}f && = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2}f && = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int\left(-D'(S)\theta l f - \psi U_{lS}\frac{l}{\theta} - \phi U_S + \nu^g\left(\frac{U_{qS}}{U_c} - \frac{U_q U_{cS}}{U_c^2}\right)\right) && = \mu
\end{aligned}
$$

Using the commodity wedge constraint $1 + \bar{\tau}^g(\theta) = \frac{U_q}{U_c}$,

$$
\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl}\frac{l}{\theta} + \nu^g\left(\frac{U_{cq}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c}\right) && = \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R}\frac{\partial V}{\partial Q}f(\theta) - \psi U_{ql}\frac{l}{\theta} + \mu S_2 f + \nu^g\left(\frac{U_{qq}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cq}}{U_c}\right) && = \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -D(S)\theta f - \phi U_l + \nu^g\left(\frac{U_{ql}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cq}}{U_c}\right) && = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi && = \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w}f && = \phi\beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R}\frac{\partial V_{t+1}}{\partial w_2}f && = -\psi\beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int\left(-D'(S)\theta l f - \psi U_{lS}\frac{l}{\theta} - \phi U_S + \nu^g\left(\frac{U_{qS}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cS}}{U_c}\right)\right) && = \mu
\end{aligned}
$$

Next, we substitute ϕ using the first FOC into the other FOCs.

$$\phi = \frac{f - \psi U_{cl}\frac{l}{\theta} + \nu^g\left(\frac{U_{cq}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c}\right)}{U_c}$$

Let's start deriving the expression for the (constrained)-optimal labor wedge, taking $\frac{\partial \mathcal{H}}{\partial q}$ as a starting point and using $1-\tau^y = -\frac{U_l}{D(\theta)\theta U_c}$.

$$\begin{aligned}
&-D(S)\theta f - \frac{f - \psi U_{cl}\frac{l}{\theta} + \nu^g\left(\frac{U_{cq}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c}\right)}{U_c}U_l + \nu^g\left(\frac{U_{ql}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cq}}{U_c}\right) = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\implies &-D(S)\theta f - \frac{U_l}{U_c}f + \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} - \nu^g\left(\frac{U_{cq}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c}\right)\frac{U_l}{U_c} + \nu^g\left(\frac{U_{ql}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cq}}{U_c}\right) = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\implies &-D(S)\theta f - \frac{U_l}{U_c}f + \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} - \nu^g\left(\frac{U_{cq}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c}\right)\frac{U_l}{U_c} + \nu^g\left(\frac{U_{ql}}{U_l} - (1+\bar{\tau}^g)\frac{U_{cq}}{U_l}\right)\frac{U_l}{U_c} = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\implies &-D(S)\theta f - \frac{U_l}{U_c}f + \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} - \nu^g\left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c} - \frac{U_{cq}}{U_l}\right)\right)\frac{U_l}{U_c} = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\implies &-D(S)\left(1 + \frac{U_l}{D(\theta)\theta U_c}\right)\theta f + \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} - \nu^g\left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c} - \frac{U_{cq}}{U_l}\right)\right)\frac{U_l}{U_c} = \frac{1}{\theta}\psi\left[U_{ll}l + U_l\right] \\
\implies &-D(S)\tau^y\theta f + \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} - \nu^g\left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c} - \frac{U_{cq}}{U_l}\right)\right)\frac{U_l}{U_c} = -\frac{1}{\theta}\psi\left[\frac{U_{ll}l + U_l}{U_l}\right](-U_l) \\
\implies &-D(S)\tau^y\theta f - \nu^g\left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c} - \frac{U_{cq}}{U_l}\right)\right)\frac{U_l}{U_c} = -\frac{1}{\theta}\psi(1+\epsilon)(-U_l) - \psi\frac{U_{cl}}{U_c}\frac{U_l l}{\theta} \\
\implies &\tau^y + \nu^g\left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c} - \frac{U_{cq}}{U_l}\right)\right)\frac{U_l}{D(S)\theta U_c}\frac{1}{f} = -\frac{1}{D(S)\theta^2 f}\psi(1+\epsilon-\gamma)(-U_l) \\
\implies &\tau^y - \nu^g\left(\frac{U_{cq}}{U_c f} - \frac{U_{ql}}{U_l f} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c f} - \frac{U_{cq}}{U_l f}\right)\right)(1-\tau^y) = \frac{(1-\tau^y)U_c}{\theta f}\psi(1+\epsilon-\gamma) \\
\implies &\frac{\tau^y}{1-\tau^y} = \frac{U_c}{\theta f}\psi(1+\epsilon-\gamma) + \nu^g\left(\frac{U_{cq}}{U_c f} - \frac{U_{ql}}{U_l f} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c f} - \frac{U_{cq}}{U_l f}\right)\right)
\end{aligned}$$

I define the following to keep notation short:

$$\Gamma_l^g \equiv \frac{U_{cq}}{U_c f} - \frac{U_{ql}}{U_l f} - (1+\bar{\tau}^g)\left(\frac{U_{cc}}{U_c f} - \frac{U_{cq}}{U_l f}\right)$$

If we isolate ψ,

$$\psi = \left[\left(\frac{\tau^y}{1-\tau^y}\right) - \nu^g\Gamma_l^g\right]\frac{\theta f}{U_c}\frac{1}{1+\epsilon-\gamma}$$

Because $\psi = -\frac{1}{\beta R}\frac{\partial V_{t+1}}{\partial w_2} f$,

$$-\frac{\partial V_{t+1}}{\partial w_2} = \beta R \left[\left(\frac{\tau^y}{1-\tau^y} \right) - \nu^g \Gamma_l^g \right] \frac{\theta}{U_c} \frac{1}{1+\epsilon-\gamma}$$

By the envelope condition,

$$\lambda_{2,t+1} = \beta R \left[\left(\frac{\tau^y_{t+1}}{1-\tau^y_{t+1}} \right) - \nu^g_{t+1} \Gamma^g_{l,t+1} \right] \frac{\theta}{U_{c,t+1}} \frac{1}{1+\epsilon_{t+1}-\gamma_{t+1}}$$

Next, we find the expression for $\psi(\theta)$ by integrating the FOC for u.

$$\dot{\psi} = -\lambda_1 f + \lambda_2 f_2 + \frac{f - \psi U_{cl}\frac{l}{\theta} + \nu^g \left(\frac{U_{cq}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c} \right)}{U_c}$$

$$\underbrace{\dot{\psi}}_{\frac{dy}{dx}} = \underbrace{-\lambda_1 f + \lambda_2 f_2 + \frac{f + \nu^g \left(\frac{U_{cq}}{U_c} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c} \right)}{U_c}}_{\text{Q}} - \underbrace{\frac{1}{\theta}\gamma}_{\text{P}} \underbrace{\psi}_{\text{y}}$$

This is a first-order differential equation of form $\frac{dy}{dx} = Q + Py$, which has the following solution:

$$\psi(\theta) = \int_\theta^\infty \exp\left[-\int_\theta^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_c(x)} \left[1 + \nu^g \Gamma_q^g \right] f(x) - \lambda_1 f(x) + \lambda_2 f_2(x) \right) dx$$

where

$$\Gamma_q^g \equiv \frac{U_{cq}}{U_c f} - (1+\bar{\tau}^g)\frac{U_{cc}}{U_c f}$$

Using the boundary condition that $\psi(0) = 0$, we can find λ_1.

$$\lambda_{1,t} = \frac{\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} \left[1 + \nu_t^g \Gamma_{q,t}^g \right] f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_t(x)}$$

We substitute ψ and $\lambda_{2,t}$ (iterating from $\lambda_{2,t+1}$) into the expression for the labor wedge. Since the steps are the same as in Golosov, Troshkin, et al. (2016) aside for some different substitutions, I skip some of the algebra.

$$
\begin{aligned}
\frac{\tau_t^y}{1-\tau_t^y} &= \int_\theta^\infty \exp\left[-\int_\theta^x \gamma(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{1}{U_{c,t}(x)}\left[1+\nu_t^g\Gamma_{q,t}^g\right]f_t(x)-\lambda_{1,t}f_t(x)+\lambda_{2,t}f_{2,t}(x)\right)dx\,\frac{U_{c,t}(\theta)}{\theta_t f_t(\theta)}(1+\epsilon_t(\theta)-\gamma_t(\theta))+\nu_t^g\Gamma_{l,} \\
&= \cdots \\
&= \int_\theta^\infty \frac{U_{c,t}(\theta)}{U_{c,t}(x)}\exp\left[-\int_\theta^x \gamma(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right](1+\epsilon_t(\theta)-\gamma_t(\theta))\left[1+\nu_t^g\Gamma_{q,t}^g\right]\frac{f_t(x)}{\theta_t f_t(\theta)}\,dx - \\
&\quad \int_\theta^\infty \frac{U_{c,t}(\theta)}{U_{c,t}(x)}\exp\left[-\int_\theta^x \gamma(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]U_{c,t}(x)(1+\epsilon_t(\theta)-\gamma_t(\theta))\lambda_{1,t}\frac{f_t(x)}{\theta_t f_t(\theta)}\,dx + \\
&\quad \beta R\left[\left(\frac{\tau_{t-1}^y}{1-\tau_{t-1}^y}\right)+\nu_t^g\Gamma_{l,t}^g\right]\frac{1+\epsilon_t(\theta)-\gamma_t(\theta)}{1+\epsilon_{t-1}(\theta)-\gamma_{t-1}(\theta)}\frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)}\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]f_{2,t}(x)\,dx}{\theta_t f_t(\theta)}+\nu_t^g\Gamma_{l,t}^g \\
&= \cdots \\
&= (1+\epsilon_t(\theta)-\gamma_t(\theta))\frac{1-F_t(\theta)}{\theta_t f_t(\theta)}\int_\theta^\infty \exp\left[\int_\theta^x\left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})}-\gamma_t(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right)d\tilde{x}\right](1+\nu_t^g\Gamma_{q,t}^g-\lambda_{1,t}U_{c,t}(x))\frac{f_t(x)dx}{1-F_t(\theta)}+ \\
&\quad \beta R\left[\left(\frac{\tau_{t-1}^y}{1-\tau_{t-1}^y}\right)+\nu_t^g\Gamma_{l,t}^g\right]\frac{1+\epsilon_t(\theta)-\gamma_t(\theta)}{1+\epsilon_{t-1}(\theta)-\gamma_{t-1}(\theta)}\frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)}\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]f_{2,t}(x)\,dx}{\theta_t f_t(\theta)}+\nu_t^g\Gamma_{l,t}^g
\end{aligned}
$$

This yields the following modified ABCD formula.

$$
\frac{\tau_t^y}{1-\tau_t^y} = A_t(\theta)B_t(\theta)C_t(\theta)+\left[\left(\frac{\tau_t^y}{1-\tau_t^y}\right)+\nu_t^g\Gamma_{q,t}^g\right]D_t(\theta)+\nu_t^g\Gamma_{l,t}^g
$$

where

$$
\begin{aligned}
A_t(\theta) &\equiv (1+\epsilon_t(\theta)-\gamma_t(\theta)) \\
B_t(\theta) &\equiv \frac{1-F_t(\theta)}{\theta_t f_t(\theta)} \\
C_t(\theta) &\equiv \int_\theta^\infty \exp\left[\int_\theta^x\left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})}-\gamma_t(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right)d\tilde{x}\right](1+\nu_t^g\Gamma_{q,t}^g-\lambda_{1,t}U_{c,t}(x))\frac{f_t(x)dx}{1-F_t(\theta)} \\
D_t(\theta) &\equiv \frac{A_t(\theta)}{A_{t-1}(\theta)}\frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)}\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]f_{2,t}(x)\,dx}{\theta_t f_t(\theta)}
\end{aligned}
$$

C.4 Constrained-optimal savings wedge (suboptimal commodity wedge)

TBA